

Aufgabe 1:

$$\text{DFT: } F(u) = \sum_{x=0}^3 f(x) \exp(-i \frac{2\pi u x}{4})$$

↳ $f(1)$ und $f(3)$ gleich 0 per Definition

$$\Rightarrow F(u) = 1 \cdot \exp(-i \frac{2\pi u \cdot 0}{4}) + 2 \cdot \exp(-i \frac{2\pi u \cdot 2}{4})$$

Eulersche
Relation
 $\Rightarrow$

$$= \underbrace{\cos(0) - i \sin(0)}_{=1} + 2 \left(\underbrace{\cos(\pi u)}_{=(-1)^u} - i \underbrace{\sin(\pi u)}_{=0} \right)$$

$$= 1 + 2(-1)^u$$

$$F(0) = 1 + 2(-1)^0 = 3$$

$$F(1) = 1 + 2(-1)^1 = -1$$

$$F(2) = 1 + 2(-1)^2 = 3$$

$$F(3) = 1 + 2(-1)^3 = -1$$

$$F = [3, -1, 3, -1]$$